

Engineering collective attention in the age of artificial intelligence

Received: 10 December 2025

Accepted: 1 March 2026

Published online: 21 March 2026

Cite this article as: Zhu R. Engineering collective attention in the age of artificial intelligence. *Discov Artif Intell* (2026). <https://doi.org/10.1007/s44163-026-01084-8>

Ruibiao Zhu

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

ARTICLE IN PRESS

Engineering Collective Attention in the Age of Artificial Intelligence

Ruibiao Zhu^{1,*}

¹School of Computing, College of Systems and Society, The Australian National University, ACT, Australia

*ruibiao.zhu@anu.edu.au

ABSTRACT

This article explores how collective attention can be both disrupted and enhanced by artificial intelligence. It examines how the rise of algorithmic recommendation systems, generative media, and large-scale language models has transformed public communication and redefined what captures human attention. The analysis identifies the dual nature of artificial intelligence: while it can distort information ecosystems through deepfakes, social bots, and engagement-driven algorithms, it also holds the potential to strengthen collective reasoning by improving access to reliable knowledge and facilitating the clarification of complex information. Drawing on interdisciplinary research, the article develops a multilevel framework for understanding and improving collective attention. At the individual level, it emphasizes education, digital literacy, and critical awareness to build cognitive resilience. At the governmental level, it assesses regulatory and ethical strategies for ensuring transparency, accountability, and fairness in the design and deployment of AI systems. At the societal level, it highlights the promise of human–AI collaboration to guide attention toward truth, empathy, and shared problem-solving. The article concludes that collective attention **might** be engineered in a beneficial way when artificial intelligence is governed transparently, used ethically, and integrated with public oversight to reinforce informed, cohesive, and resilient democracies.

1 What is collective attention?

Collective attention is a situationally informed mental state in which an agent represents the world from a first-person-plural point of view—we are attending to X—brought about by the explicit or implied co-attention of one’s social group. It is not restricted to physical co-presence and can arise when co-attention is observed, communicated, imagined, or situationally implied (e.g., mass media events). This “we-mode” distinguishes collective attention from mere awareness that others share one’s beliefs or attitudes; it concerns concurrent attention to the same stimulus and can be accurate or mistaken depending on whether co-attendants in fact align. The theory of shared/collective attention makes the definitional core explicit: “the perception of in-the-moment attention to an object from a first-person-plural perspective.” This is the minimal representational shift from I am attending to we are attending [1].

Developmental work on joint attention shows how infants coordinate gaze and gestures to share refer-



ence, scaffolding later social cognition and learning [2]. Joint attention typically involves dyadic/triadic gaze and overt signaling (e.g., pointing), whereas collective attention, as used here, does not require eye contact or gaze-following: individuals can be collectively attending without looking at each other, or even without co-presence, so long as the first-person-plural stance is activated [3]. At the same time, classic joint-attention skills highlight why collective attention is powerful: early sensitivity to others' attention is linked to deeper encoding, language growth, and social competence [2].

Collective attention also bears on common knowledge—the recursive state of “everyone knows that everyone knows . . .”. Philosophical accounts hold that common knowledge typically arises when information is received publicly and simultaneously, conditions under which the awareness that “we are attending” is especially likely [1]. In this view, collective attention is an economical indicator of what is (likely to be) common knowledge: because it represents a unified collective perspective, it reduces the need for higher-order “I know that you know” computations (cognitive frugality argument) [1].

The representational mechanism is deliberately minimal: a dyadic relation between the collective self and the attended target ($\text{we} \rightarrow \text{X}$), rather than triadic mind-reading about what others think of one's attention. This simplicity makes it both fast and broadly applicable (e.g., anonymous co-viewing of a broadcast), while still channeling processing to co-attended content [1, 4]. Consistent with developmental evidence that joint attention recruits anterior and posterior attention systems, the collective-attention account predicts an attentional gain on shared targets rather than obligatory mentalizing about partners' inner states [2].

Two recurrent situational features strengthen the state: (a) simultaneity and (b) relational proximity (in-groupness). Simultaneity ensures that co-attendants are likely to encode the same elements at the same time; in-groupness signals prospects for future coordination, encouraging a “we” representation [1, 4].

Collective attention is stronger (and more consequential) when there is clear evidence or expectation of simultaneous co-attention with in-group others; it is weaker with relationally distant others or asynchronous exposure [1, 4]. Minimal social ties are often sufficient to evoke the state, but clinical and individual-difference work suggests limits: for example, individuals on the autism spectrum may not show the typical learning boost under collective attention [2]. Finally, like any mental state, collective attention can be mistaken—people may believe “we are attending” when co-attendants are not in fact aligned.

When a stimulus is valenced, shared attention intensifies the corresponding emotion. In laboratory and field paradigms, people like pleasant tastes more—and dislike unpleasant tastes more—when tasting simultaneously with another person (even without communication), an effect attributed to shared attention [5]. When people observe a behavior under conditions of simultaneous in-group co-attention, they are more likely to adopt it. In two experiments, participants imitated the format of a conversational exchange more strongly when they believed a similar other was simultaneously observing the same exchange—documenting “shared in-group attention” as a uniquely potent social-learning context [6]. Converging cognitive-neuroscience evidence indicates that shared attention amplifies neural processing of co-attended information, consistent with the first-person-plural account [7]. By prioritizing the same

information for co-attendants, collective attention produces cognitive alignment. This alignment makes it more likely that the alignment itself becomes an object of attention (“we are attending to our shared state”), supporting the representation of common knowledge, which is a key driver of coordination and collective action [1].

Collective attention is a first-person-plural attentional stance that (i) is distinct from joint attention and shared beliefs, (ii) emerges reliably under simultaneous co-attention with in-group others, and (iii) amplifies cognitive processing, motivation, affect, and learning for the attended target. Through this amplification—and its coupling with public, simultaneous exposure—collective attention both indicates and helps produce the common knowledge that underwrites human coordination. In everyday interaction, it is tractable in behavior and physiology, and in group contexts it functions as a cognitive lever for shared understanding and collective action [1, 4].

2 How can collective attention be problematic?

2.1 Misinformation, polarization, and emotional amplification

Across large-scale social platforms, collective attention is not a neutral resource: it is structured by network homophily, algorithmic curation, and human affective responses that jointly bias what is seen, believed, and shared. Three failure modes recur in the corpus: (i) the outcompetition of accurate information by falsehoods, (ii) the segregation and even backfire of cross-cutting exposure, and (iii) the preferential amplification of moralized, high-arousal emotion [8, 9].

A comprehensive analysis of around 126,000 rumor cascades on Twitter demonstrates that false news reliably outperforms true news across the main axes of diffusion [8]. Falsehoods reach more people (the top 1% of false cascades routinely reached 1,000–100,000 users while truth rarely exceeded 1,000), travel deeper (top false cascades exceeded 19 hops), spread faster (e.g., reaching 1,500 people 6 times faster), and exhibit greater structural virality than truths; these patterns are especially pronounced for political content. Moreover, the advantage of falsehood is not explained by network position or user status (spreaders of false news had fewer followers, were less active, and were less often verified) and persisted after removing bot traffic; bots accelerated both true and false news at roughly the same rate, implying the critical driver is human sharing behavior. False stories were also measurably more novel (higher information uniqueness and K–L divergence), and they elicited different emotions—replies to false news contained more surprise and disgust, whereas replies to true news contained more anticipation, joy, and trust. These findings explain why collective attention can be systematically captured by misinformation: novelty and affect give falsehoods a sharing advantage even when network and account features favor truth [10, 11, 12].

Platform architectures channel attention through homophilous networks and user choices in ways that limit cross-cutting information and can exacerbate division. Using behavioral data from 10.1 million U.S. Facebook users, isolating three stages—what friends share, what the News Feed shows, and what users click [9]. Cross-ideological content declines at each step: algorithmic ranking slightly reduces exposure to cross-cutting links, but individual selection plays the stronger role in limiting actual consumption.

Despite nontrivial cross-party ties (median cross-ideological friendship shares 0.20 for liberals and 0.18 for conservatives), users ultimately click a markedly smaller share of discordant news than they could have seen. The authors conclude that, relative to algorithms, user choices are the primary constraint on cross-cutting exposure on Facebook.

At the ecosystem level, comparative analyses reveal that homophilic clustering and like-minded diffusion dominate across platforms, with platform design moderating its strength. Facebook and Reddit are directly compared and showed higher segregation on Facebook: users' "influence sets" remain closer to seed ideology, community clusters are more ideologically aligned, and simulated information spreading preferentially reaches like-minded users [9]. A "clear-cut distinction" emerges between social media whose feeds are less user-tweakable (e.g., Facebook, Twitter) and those with user-tunable feeds (e.g., Reddit), with the former displaying stronger echo-chamber signatures.

Crucially, even well-intentioned attempts to increase cross-cutting exposure can backfire. In a field experiment, Republicans randomly incentivized to follow a liberal Twitter bot became more conservative over a month; Democrats showed a small, statistically non-significant liberal shift when following a conservative bot. The authors caution that interventions that "push" opposing content into people's feeds may be counterproductive in some contexts [13].

Collective attention also privileges content framed with moral emotion. An analysis of 563,312 tweets about gun control, same-sex marriage, and climate change shows that each additional moral-emotional word in a message increases its predicted retweet count by about 20% [14]. This "moral contagion" is bounded by group identity: moral-emotional language boosts diffusion more within ideological in-groups than across them. The result highlights a mechanism by which moralized discourse can intensify within echo chambers and contribute to ideological sorting.

The preference for high-arousal negative affect further skews attention. In Weibo interaction networks, anger shows significantly stronger correlation between connected users than joy (with sadness comparatively weak), and the correlation increases with tie strength and degree—evidence that anger travels more readily through social ties than other affects [15]. This pattern was linked to real-world grievance-laden topics (e.g., corruption, food safety), clarifying why anger-laden content can dominate collective attention during contentious events.

The recent work connects these affective dynamics to platform feedback and norms. Across 12.7 million tweets and two behavioral experiments, moral outrage expression increases with positive social feedback (likes/retweets), consistent with reinforcement learning. At the same time, users conform to expressive norms in their networks; in ideologically extreme networks—where outrage is common—users rely less on feedback because norms already signal what is appropriate to express. The authors argue that platform design can modulate these learning signals by shaping visibility and feedback, thereby influencing how outrage scales within networks [16].

When collective attention is routed through homophilous networks and optimized for novelty and engagement, falsehoods gain structural and psychological advantages over truths, moralized content out-

competes neutral information, and cross-cutting exposure is both scarce and sometimes counterproductive. False news' novelty and fear–disgust–surprise profile make it more shareable at the very moment when users are incentivized—by peers and platforms—to produce attention-grabbing content; this is particularly acute in political domains. The same environments reinforce anger and outrage through social feedback and norms, while homophily and user selection concentrate attention within like-minded clusters and limit corrective scrutiny. Consequently, collective attention can become a runaway amplifier for misinformation, ideological segregation, and emotional extremity—even when many users have cross-party ties or platforms remove bots—because the core drivers are human sharing choices interacting with network structure and feedback design [8, 9, 16].

2.2 Agenda volatility, limited attention and public risk

The dynamics of collective attention—its volatility, scarcity, and amplification—are central to understanding how public priorities and risk perceptions evolve. Collective attention operates under constraints analogous to a finite resource: as new issues compete for notice, older ones are displaced, leading to unstable agendas, uneven learning, and distorted responses to emerging risks [17, 18, 19, 20].

Public and policy agendas rarely evolve smoothly. The “Issue-attention cycle” describes how social problems move through stages of alarmed discovery, enthusiasm, and eventual decline as the public's attention wanes once costs of sustained engagement become evident. This cyclical rise and fall creates volatility: even urgent problems fade from the spotlight not because they are solved, but because the collective capacity to attend to them is exhausted [20].

At the institutional level, such volatility arises from positive feedback loops between issue framing and venue change. When new images of a problem—such as environmental risk—gain traction, they mobilize new actors and shift policymaking venues. This produces punctuated shifts in policy, alternating long periods of stability with abrupt bursts of change. Collective attention is therefore not simply variable; it is structurally unstable, driven by feedback between cognitive focus and institutional response [21].

Quantitative evidence shows that this volatility is accelerating. Analyzing data from social media, books, films, and scientific publications, they show that the duration of collective attention to cultural items has shortened markedly over recent decades, while turnover rates have increased. The model—rooted in limited cognitive capacity and competition among topics—suggests that rising volumes of information production and consumption compress the life cycles of public attention, creating increasingly fragmented and transient collective focus [18].

Collective attention, like individual attention, is finite. A wealth of information creates a poverty of attention [17]. Subsequent research on information dynamics confirms that when content proliferates, attention becomes the limiting factor in the social processing of information. This competition has been formalized: as content production (imitation) and consumption (saturation) rates rise, topics peak and decline more quickly, producing “temporal fragmentation” of public discourse. The result is an attention economy characterized by constant churn and shallow engagement [18].

This scarcity induces systemic distortions in risk communication. In the case of Google Flu Trends, big-data systems that depend on collective signals are themselves subject to the instabilities of public attention. Media coverage and algorithmic feedback can overamplify seasonal search behaviors, producing “big data hubris” and false inferences about public health threats. In such settings, attention itself becomes an endogenous variable—shaped by media dynamics and platform algorithms—introducing volatility into risk assessment [22].

Collective attention does not merely fluctuate; it magnifies or dampens perceptions of danger. The framework links technical assessments of hazards with psychological and cultural processes that determine public response. Information about risks passes through “amplification stations”—scientists, media, and social networks—that filter, dramatize, or contest signals, producing waves of amplification or attenuation. Amplified risks trigger behavioral and institutional reactions—market shocks, regulatory overcorrections, or stigmatization—whose secondary effects may exceed the original hazard. Conversely, attenuated risks, such as chronic environmental exposures, may be dangerously neglected [19].

This dynamic connects closely with positive-feedback models: both describe how new information or images can trigger disproportionate shifts in public and institutional behavior [19]. When collective attention is volatile and bounded, minor cues can cascade into major policy responses, while sustained engagement with complex, long-term risks—climate change, pandemics, or financial fragility—is undermined by attentional fatigue [17, 20].

Together, these perspectives reveal how limited collective attention generates instability and public risk in three intertwined ways:

- Temporal Compression — As attention cycles accelerate, complex problems are addressed through fleeting surges of interest followed by neglect. This undermines long-term policy learning and preparedness [18, 20].
- Selective Amplification — Media and network effects amplify vivid, novel, or emotionally charged hazards, while attenuating slow-moving or diffuse threats [14, 19, 23].
- Algorithmic Feedback — Automated information systems (e.g., search engines, social media feeds) reinforce and distort public focus, translating collective attention biases into misinformed risk assessments [22, 24, 25, 26].

In this sense, agenda volatility and limited attention are not just epistemic limitations but risk multipliers. They shape the social distribution of concern, the allocation of institutional capacity, and the emotional tone of public discourse [19, 20].

2.3 Herding, cascades, and market distortions

Collective attention can become problematic in markets by inducing herding—agents mimic predecessors—and informational cascades—agents optimally ignore their private signals and follow the crowd.

Two canonical models make the logic transparent. In the sequential-choice model, once a single option has been chosen by two people, rational late movers optimally follow that option even if their private signals disagree—a herd externality in which each agent’s mimicking choice deprives later agents of information. Because herding dampens the information content of actions, the equilibrium can be inefficient; ex-ante welfare may be higher if early agents cannot observe others, and the probability that nobody ever chooses the correct option is bounded away from zero (and can be large when private signals are weak). The model also predicts volatile, path-dependent outcomes because early, partly random signals set the direction of the herd [27].

Informational cascades is formalized in which actions “speak louder than words”: as soon as the public record of predecessors’ actions outweighs an individual’s private signal, it is optimal to ignore that signal and conform [28]. Cascades are fragile: small public disclosures can overturn them, and even with high signal precision the ex-ante probability of an incorrect cascade is non-trivial. The model also yields clear policy levers: timely public signals can break bad cascades; conversely, observing only distributions of choices (not order) can still sustain cascades. These informational mechanisms coexist with reputational motives for correlated behavior. In the agency model, managers rationally mimic others to protect reputation, even when doing so means discarding valuable private information, generating socially inefficient clustering of investment decisions [29].

In retail segments of equity markets, attention shocks create predictable, temporary price pressure. Using Google search volume as a direct attention measure, weekly spikes in ticker searches lead other attention proxies and predict positive abnormal returns over the next two weeks, followed by reversal over the subsequent year [30]. Effects are stronger for small-cap stocks and for stocks more heavily traded by retail investors (interactions with market-center measures based on SEC rules), and the trading link is especially pronounced at venues catering to less sophisticated retail order flow. These patterns are precisely what attention-based herding models predict: when many investors simultaneously look at the same names, buying pressure exceeds selling pressure (few retail investors short), pushing prices away from fundamentals before they mean-revert as attention dissipates. Independent evidence points to the same mechanism. Individuals are net buyers of “attention-grabbing” stocks—those in the news, with extreme returns, or unusually high volume—because purchase decisions require search over many alternatives, whereas sales are constrained to the investor’s existing holdings [31]. The result is temporary price pressure and subsequent reversal, an attention-driven distortion of prices.

Attention-induced cascades are particularly visible at initial public offerings (IPOs). A sharp, transitory surge in Google searches around IPO dates [30]. IPOs with higher pre-issue search exhibit significantly larger first-day returns increase in cross-sectional regressions controlling for standard IPO determinants and stronger long-run underperformance when initial returns are high. Consistent with the cascade logic, broad retail attention can swing demand toward newly listed names, generating price pops that later deflate as attention recedes and private information reasserts itself. Actions are not sufficient statistics for private signals, so the public history loses informational content once herding starts. The probability

that no one ever chooses the correct option remains strictly positive regardless of sample size, and that suppressing early observability can raise welfare because it prevents premature herds [27]. Also, the cascades are idiosyncratic and fragile, so small shocks—public announcements, one salient deviation—can flip collective behavior, creating excess volatility relative to fundamentals [28]. In practice, the combination of (a) attention shocks that synchronize trades and (b) fragile cascades that can reverse suddenly helps explain short-horizon momentum followed by reversals and episodic booms and busts—classic signatures of market distortion. When reputational forces dominate, managers herd on investment choices—even when private information points elsewhere—producing correlated errors and misallocation of capital. Because the behavior is privately rational but socially costly, it exemplifies a distinct route from collective attention to distortion.

Collective attention is not merely a background condition; it is an equilibrium force. When agents rationally infer from others' actions, herds and cascades arise that can suppress private information, move prices away from fundamentals, and misallocate capital. These dynamics are theoretically well understood—through informational and reputational mechanisms—and empirically visible in the predictable price pressure and reversals linked to attention shocks and in IPO waves. Recognizing the conditions that foster cascades and designing information environments that keep early signals from being overweighted are central to limiting attention-driven market distortions [27, 28].

2.4 Collective attention in AI systems

Collective attention – the aggregate pattern of what information people focus on – is increasingly mediated by AI-driven platforms and algorithms. While these systems help direct attention, they can also introduce problematic feedback loops that distort information ecosystems. [Classical accounts of propaganda and consent-manufacture emphasize that modern publics can be steered through the strategic control of information, symbols, and salience \[32, 33, 34, 35, 36\]. The digital revolution changes this project along dimensions: \(i\) frequency \(continuous, real-time optimization rather than episodic campaigns\), \(ii\) scale \(automated delivery to billions\), \(iii\) personalization \(data-driven segmentation and micro-targeting\), and \(iv\) manipulation capacity \(dynamic, individualized choice-architecture interventions sometimes described as hypernudges or big nudging\), and \(v\) *defense asymmetry* \(generative AI makes mis- and disinformation cheap to produce at scale, while detection and verification remain limited and brittle\) \[37, 38, 39, 40, 41, 42\]. Both empirical audits and theoretical models reveal multiple dimensions in which collective attention, when filtered through engagement-optimized AI or large language models \(LLMs\), may lead to adverse outcomes \[43\]. Below, I analyze how biased or constrained attention can amplify polarizing content, enable low-quality information to thrive, distort user preferences, and even erode the knowledge base of AI models.](#)

[A key \(and often under-discussed\) mechanism is the auctioning infrastructure that monetizes attention in the digital ad market. Data privacy is endangered by the vast data broadcasts inherent in real-time bidding \(RTB\), prompting legal scholars to warn that current ad-tech practices are largely incompatible](#)

with privacy laws and may expose sensitive information [44, 45]. Foreign influence operations have found fertile ground in micro-targeted advertising, with studies and official investigations documenting how hostile actors can exploit major ad platforms to amplify disinformation and undermine democratic processes [38, 46]. Meanwhile, the infrastructure vulnerabilities in the ad ecosystem—from malware-laden ads (malvertising) to fraud schemes—pose cyber risks that can affect economic and security interests alike [47, 48].

Social media platforms that rank posts based on engagement tend to bias collective attention toward divisive or emotionally charged content. In a preregistered audit of Twitter’s algorithm, researchers found that the engagement-based ranking amplifies emotionally charged, out-group hostile political content relative to a chronological feed, content which users reported actually made them feel worse about the opposing political group [49]. Notably, the very tweets that the algorithm elevated were not those users preferred seeing – the algorithm’s selections underperformed in satisfying users’ stated content preferences. This suggests that optimizing for clicks and reactions exploits human attentional biases at the expense of user well-being. Indeed, prior work has hypothesized that content ranking algorithms “systematically exploit human social-learning biases” – favoring posts that are prestigious, in-group oriented, moral, or emotional (so-called “PRIME” content) – in order to capture attention. By maximizing reactive engagement, algorithms may lean into our bias for sensational information, thereby amplifying polarizing material that grabs attention but is misaligned with reflective values or balanced discourse [50]. Empirical evidence of algorithmic amplification bias underscores this concern: a large-scale experiment showed that Twitter’s personalized feed gave significantly greater exposure to one side of the political spectrum (mainstream right-wing content) over the other in six out of seven countries studied [25]. In short, engagement-optimized AI platforms can skew collective attention toward partisan or provocative content, potentially exacerbating polarization and bias in public discourse.

The dynamics of collective attention are constrained by human cognitive limits, which can allow low-quality or misleading information to outcompete truth in online environments. Our attention is a finite resource, and ideas (or “memes”) must compete for this scarce attention on social networks. Agent-based models of information spread have demonstrated that even without assuming any intrinsic quality differences, this competition produces a “winner-takes-all” pattern in which a few memes go viral while most receive little attention [51]. The skewed popularity distribution observed on platforms (a handful of items achieve massive popularity) can thus emerge purely from network effects and limited attention, rather than from superior merit. When content quality is taken into account, the picture remains worrying. Simulation studies indicate that information overload and finite attention capacity degrade the system’s ability to discriminate quality [52]. As the number of items and noise in the feed grow, attention gets spread thin and the “marketplace of ideas” loses its filtering power. This social network model with real-world parameters and found that under realistic high-load conditions there is only a weak correlation between content quality and its popularity. In fact, their model predicts that low-quality information can be just as likely to go viral as high-quality information, providing a compelling explanation for the prevalence

of misinformation in online media. In essence, when collective attention is fragmented and overwhelmed, engaging or novel content can dominate regardless of accuracy. This attention economy problem enables the spread of rumors, fake news, and other misleading memes that would likely fail to gain traction in a less crowded, more quality-sensitive information environment.

More broadly, the attention economy has incentives to keep people overloaded and reactive. Classic accounts of the attention economy emphasize that as information becomes abundant, attention becomes the limiting resource and can be systematically commodified [17, 53]. Under conditions of chronic overload, individuals are pushed toward fast, reflexive consumption of whatever is most salient in the feed, increasing reliance on defaults and suggestions rather than on deliberation [37, 52]. This raises a normative warning: collective attention should not be treated as an unprotected public good that can be harvested by any actor—political or commercial—who manages to capture it [40, 53]. A concrete (and explicitly heuristic) policy principle is to preserve the bulk of citizens’ attentional capacity for self-directed life management and co-creation: for instance, limiting attention-harvesting political applications to roughly 10% and commercial applications to roughly 10%, leaving around 80% available for work, care, learning, creativity, and civic association. Protecting this “attentional commons” is also an innovation policy: societies need conditions in which novel ideas can surface and spread—even when they challenge established approaches, business interests, or political goals—rather than being crowded out by engagement-optimized incumbents [51, 54].

Another dimension of problematic collective attention arises from feedback loops in recommender systems and other AI personalization. Algorithmic confounding occurs when models are trained and updated on data that has itself been shaped by prior model recommendations [24]. This creates a self-reinforcing cycle: the algorithm learns from user behavior that it influenced, potentially narrowing the range of content it shows in the future. Simulations show that such feedback loops drive increasing homogenization of user behavior without corresponding gains in utility. Over successive recommendation cycles, users converge toward consuming a more similar set of items (reduced diversity), yet this does not improve their satisfaction or the accuracy of predictions. In fact, algorithmic feedback tends to concentrate attention on a smaller subset of content: the influence of recommendations amplifies the popularity of already popular items, skewing the distribution of content consumption toward less diverse outcomes. This means collective attention becomes more uniform and focused on what the algorithm pushes, even if that does not reflect users’ true interests [55, 56, 57].

The phenomenon can be understood through the lens of performative feedback loops in AI. When predictions or recommendations influence user choices, they effectively shape the ground truth they aim to predict, as highlighted by the performative prediction framework [58]. For example, if a recommendation system keeps suggesting a particular genre of content, users’ expressed preferences may shift toward that genre simply due to repeated exposure. Over time, the system and the users co-evolve in a closed loop. The model is optimizing for a moving target of its own making, which can entrench certain preferences or filter bubbles. User audits confirm that engagement-based rankings can diverge from what users actually

value, underscoring how these AI-driven attention loops may reduce overall welfare [49]. In summary, algorithmic feedback confounding can distort collective preferences and reduce the diversity and utility of content exposure, as the system amplifies its own biases in a performative cycle.

Collective attention problems are not limited to social media and recommender systems; they also loom in the training of AI models themselves. LLMs and other generative models learn from vast corpora of text or images that represent the accumulation of human knowledge and communication – effectively the collective content produced by society. However, as AI-generated text and images become more prevalent online, future models may end up training on data that increasingly consists of their own predecessors’ outputs rather than solely human-generated examples. Recent theoretical and empirical work indicates that naive reliance on such recycled data can trigger a degenerative process known as “model collapse” [59, 60, 61]. If an AI is repeatedly retrained on content produced by earlier models, the tails of the original data distribution gradually disappear from the model’s learned representation. In other words, the model begins to forget or misrepresent the true distribution of real-world data. Their analysis found that over successive generations, models lose information about infrequent or nuanced events, and their outputs converge to an overly simplified distribution with much lower variance [59, 60, 61]. This collapse is not easily reversible; it represents an erosion of the model’s epistemic foundation as it drifts from reality with each feedback cycle. Essentially, the collective knowledge encoded in AI systems can degrade when the system’s own outputs start to dominate the training pool – the AI becomes a self-referential echo chamber [60, 61]. Maintaining epistemic integrity in LLMs, therefore, requires caution against uncontrolled recursive training on AI-generated content. The findings imply that sustaining the diversity and accuracy of an AI’s knowledge base will depend on continually grounding training in genuine human data and experiences. Without such intervention, collective attention as captured by AI models may increasingly reflect a distorted, collapsing version of reality, undermining the long-term utility and trustworthiness of AI systems.

3 How can collective attention be engineered—controlled and improved?

In contemporary digital societies, collective attention—the distributed and aggregated focus of human cognition across topics and events—has become one of the most influential yet vulnerable social phenomena. As information ecosystems expand exponentially, individuals and communities increasingly rely on algorithmic mediation to determine what deserves attention [62, 63]. Artificial intelligence (AI), especially large language models (LLMs), has emerged as a key actor in this process, shaping the circulation, visibility, and prioritization of information at global scale [64, 65].

While AI systems can assist in organizing knowledge, they also risk amplifying bias, disinformation, and polarization [43, 66]. Collective attention can therefore be “engineered”—either intentionally or inadvertently—through algorithmic design, data training, and content recommendation. The challenge lies in directing this capacity toward socially constructive ends, [using governance frameworks that operationalize human rights, transparency, and risk management \(e.g., the OECD AI principles, UNESCO’s](#)

Recommendation on the Ethics of AI, NIST's AI Risk Management Framework, and the U.S. Blueprint for an AI Bill of Rights) [67, 68, 69, 70]. This section analyzes how AI and LLMs influence collective attention, explores the dangers of unregulated manipulation such as election-related deepfakes and algorithmic amplification, and identifies methods to enhance attention integrity through education, regulation, and collaborative human–AI frameworks.

3.1 The transformative influence of AI on collective attention

The attention economy has evolved into a complex feedback system driven by algorithmic selection. Social media and news platforms employ recommendation engines that learn from engagement metrics—clicks, shares, or watch time—to predict and prioritize content likely to maintain user interest [26]. This optimization process creates a self-reinforcing cycle: emotionally charged, controversial, or sensational material tends to attract more engagement and is therefore more frequently promoted [14, 50]. Over time, this mechanism can distort the public sphere by amplifying extreme opinions, fostering echo chambers, and marginalizing moderate or nuanced discourse [9, 13].

Crucially, AI systems do not merely route attention; they increasingly participate in the epistemic processes that constitute collective attention itself. Ranking, summarization, and conversational answering can frame what counts as relevant evidence, shape credibility judgments, and steer which interpretations appear reasonable. These are inherently value-laden decisions (about what to optimize, which risks to trade off, and whose interests to privilege), and the risks of downstream harm from language models include precisely such representational and allocative effects on beliefs and decision-making [65, 71]. Experimental evidence also suggests that even subtle source cues in AI outputs can shift users' trust and inferences—effectively making the model an epistemic gatekeeper rather than a neutral conduit [72].

Empirical studies reveal that algorithmic curation has not consistently strengthened democratic participation or informed deliberation. Instead, it often fragments public attention into competing ideological clusters [73]. The incentives embedded in machine-learning systems encourage the viral spread of divisive material rather than the slow exchange of reasoned argument. Consequently, large segments of the population may experience different mediated realities depending on which platforms or feeds they consume.

A particularly striking manifestation of engineered attention is the rise of synthetic media. Advances in generative AI have enabled the production of hyper-realistic audio and video fabrications—commonly called deepfakes—that can manipulate perception with minimal technical effort [74]. During Slovakia's 2023 national election, for example, an AI-generated audio recording circulated widely online, falsely suggesting that opposition leaders were plotting to falsify results. The timing of this release, coinciding with an official media blackout, demonstrated how fabricated material can dominate public conversation during moments of political sensitivity. Although the election outcome also reflected existing societal divisions, the incident illustrated how generative AI can redirect collective focus toward deceptive narratives in a matter of hours [75, 76].

Automated accounts powered by AI add another dimension to this challenge. Bot networks are capable of flooding social spaces with repetitive messages, inflating the perceived popularity of fringe views, or steering trending topics to favor specific agendas [77, 78, 79]. Because these systems exploit attention-based ranking, they can crowd out authentic discourse and distort what communities believe others are discussing [80]. This coordinated manipulation highlights the vulnerability of collective attention when informational supply far exceeds human capacity for verification. In the absence of deliberate countermeasures, AI-driven communication infrastructures risk converting attention into a resource that can be purchased, hijacked, or weaponized.

3.2 Individual-level resilience and cognitive empowerment

Protecting collective attention begins with strengthening individual competence. When citizens possess the skills to navigate algorithmic environments critically, the overall resilience of society improves. Digital literacy, critical thinking, and psychological preparedness are thus fundamental to reducing susceptibility to manipulation. Contemporary research in cognitive and behavioral science indicates that simple, evidence-based interventions can significantly improve information discernment. Short educational modules that explain common misinformation strategies (e.g., emotional manipulation and impersonation) can measurably improve users' ability to detect misleading content [81, 82].

Training programs based on the principle of “pre-exposure inoculation” are particularly effective. By familiarizing people with weakened examples of misleading tactics before they encounter real ones, these exercises build cognitive resistance similar to vaccination [83]. Other educational approaches teach lateral reading—checking multiple independent sources before accepting a claim—a strategy that has been shown to improve fact-checking performance in instructional settings [84].

National campaigns have begun translating these insights into practice. In Australia, for instance, the Fighting Scams initiative launched in 2025 sought to equip citizens with practical habits for online safety. The program emphasized a simple sequence: pause before acting, confirm the legitimacy of requests, and take corrective action if a scam is suspected. Such large-scale awareness drives demonstrate how governments can cultivate a population less vulnerable to fraudulent or manipulative digital content, and research suggests that well-designed awareness campaigns can reduce financial-fraud victimization [85, 86]. Comparable efforts in other regions have focused on identifying AI-generated imagery, synthetic voices, and persuasive deepfakes. Collectively, these programs aim to sustain public trust by teaching individuals to maintain critical distance from attention-grabbing falsehoods.

Beyond formal education, design interventions at the interface level can support attentional self-regulation. Tools that display content provenance, signal the reliability of sources, or allow users to adjust algorithmic recommendations according to their values give people greater agency in managing their informational diets [87, 88]. Some social networks have integrated community-based fact-checking and contextual labeling to guide readers toward verified information without resorting to heavy-handed censorship [89, 90, 91]. Subtle design cues promoting accuracy can reduce the spread of false information

and redirect user attention toward credible sources [92, 93].

In short, the improvement of collective attention at the personal level depends on fostering awareness, critical engagement, and the capacity to identify manipulation in real time [82, 83, 84, 92]. When individuals exercise informed skepticism and employ available cognitive tools, the population as a whole becomes less reactive and more reflective—a prerequisite for balanced public attention.

3.3 Engineering collective attention at the societal Level

While individual literacy forms the foundation of attention integrity, systemic solutions are necessary to correct distortions at scale. Societal engineering of collective attention involves aligning technological infrastructures, media institutions, and public policies with normative goals such as truthfulness, diversity, and social cohesion [37, 94, 95, 96].

AI may be both, culprit and cure, depending how it is being used [43, 66]. The same generative systems that create fabricated stories can also be used to detect and mitigate them. LLMs can automatically cross-reference statements against trusted databases, highlight inconsistencies, and flag potentially misleading material for human review. Automated fact-checking have accelerated the identification of viral falsehoods, allowing journalists and researchers to respond before misinformation achieves mass circulation. Similarly, AI summarization tools can transform complex documents—scientific reports, policy drafts, or legal rulings—into accessible explanations, thereby directing attention to substantive information that might otherwise remain obscure. When guided by transparent criteria, these technologies can amplify verified knowledge rather than sensational rumor.

Technology companies, such as social media companies, hold particular responsibility, as their algorithms mediate vast shares of public focus. Redesigning recommendation systems to optimize for informational value instead of raw engagement could significantly improve attention quality [88, 95]. Adjusting ranking criteria to include social benefit metrics—such as civic relevance or factual accuracy—would reorient the attention economy toward long-term public interest [88, 95]. Another promising technique involves temporary throttling of unverified content once rapid virality is detected, giving human moderators or automated fact-checkers time to assess its reliability [97]. These algorithmic adjustments transform the invisible architecture of information flow, steering collective attention toward evidence-based material.

Governmental regulation complements technological design. Some states have implemented mandatory labeling systems for AI-generated media. For example, China's Cyberspace Administration of China and other agencies jointly released the *Measures for the Identification of Artificial Intelligence-Generated and Synthetic Content* in March 2025, which enter into force on 1 September 2025 and require both explicit user-facing labels and implicit (metadata-embedded) identifiers for AI-generated or synthetic content [98]. In Europe, the EU Artificial Intelligence Act (Regulation (EU) 2024/1689) introduces transparency obligations, including marking AI-generated outputs as machine-readable and disclosing deepfakes or AI-generated text published to inform the public on matters of public interest [99]. In North America,

similar disclosure approaches are emerging in election law; for instance, California's AB 2355 (Statutes of 2024, Chapter 260) adds disclaimer requirements for political advertisements that use artificial intelligence to generate or materially alter content [100]. The intention is not to suppress innovation but to preserve transparency in an environment where visual evidence can no longer be assumed genuine.

Public-sector communication strategies further shape collective awareness. Regular information campaigns on scams, data privacy, and media manipulation frame these issues as shared societal challenges. When coordinated across schools, workplaces, and local organizations, such programs create a culture of attentional responsibility. Empirical evaluations of these efforts, such as Australia's anti-scam initiatives, show measurable reductions in victimization rates following sustained public education [85, 86]. At scale, awareness building operates as a collective "immune system," limiting the contagion of misinformation and maintaining focus on verified concerns [83].

3.4 Transparency and regulation as mechanisms of attention governance

Ensuring that AI influences collective attention in beneficial ways requires transparent governance and proportional regulation [66]. Transparency enables external oversight and builds public trust by allowing independent verification of how AI systems function; it is also a prerequisite for meaningful algorithmic accountability when platforms mediate core functions of the public and political sphere [101, 102]. Open-source models, where code and data are available for inspection, offer one pathway. Public scrutiny exposes potential biases or imbalances in training data that might otherwise remain hidden, allowing rapid correction through community collaboration. Transparent disclosure also includes informing users when they interact with AI-generated content or when recommendations are algorithmically curated. Because opacity can be strategically produced (e.g., through technical complexity, trade secrecy, and organizational incentives), relying on individual users to "figure it out" via critical reasoning is often unrealistic [103]. Platforms that systematically shape collective attention therefore require stronger transparency, responsibility, scrutiny, and accountability than current ad hoc industry practices typically provide [69, 104].

However, openness alone cannot guarantee responsible behavior, particularly when commercial or political incentives encourage secrecy. Carefully crafted regulation is therefore essential. Increasingly, regulators are converging on risk-based and proportionate approaches that pair baseline transparency duties with stronger obligations for higher-risk uses. For example, the EU Artificial Intelligence Act establishes a tiered, risk-based regulatory structure and includes specific transparency obligations for AI-generated or manipulated content [99]. At the international level, the Council of Europe's *Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law* provides a legally binding treaty framework anchored in human-rights compatibility across the AI lifecycle [105]. Complementing these hard-law and treaty approaches, the G7 Hiroshima Process articulates voluntary guiding principles for advanced AI systems intended to support safety, security, and trustworthiness across design, development, deployment, and use [106]. In the United States, federal policy has also used executive action to steer

safety and accountability expectations across agencies and industry [107]. This need for convergence in governance becomes sharper as AI is increasingly fused with other sensing and intervention technologies (e.g., ubiquitous biometrics, affective computing, and emerging neurotechnologies). In such settings, influence can operate below the threshold of individual awareness—shaping bodily states, emotions, and thinking in ways that are hard to attribute, individually detect, or counter with digital literacy and critical reasoning alone. This creates a collective-action problem: even well-prepared individuals may be unable to recognize or resist population-scale, personalized manipulation, underscoring why interoperable international norms and coordinated regulation are needed [105]. Effective frameworks balance innovation with precaution: they restrict harmful practices, such as deploying AI for mass deception or targeted harassment, while avoiding measures that would unduly stifle technological progress. Overly rigid regulation risks freezing beneficial development, whereas flexible, adaptive oversight encourages experimentation within ethical boundaries.

This point becomes especially salient as AI is increasingly deployed not merely to provide services, but to run population-scale interventions that shape people's thoughts, emotions, and behaviors. Smart-city deployments, for example, often function as real-world testbeds in which behavioral “nudges” (e.g., changes in information presentation, default options, or urban interfaces) are evaluated in situ. In such settings, the ethical boundary between policy implementation and human-subject experimentation can blur, raising the need to apply established psychological research-ethics safeguards—including risk proportionality, independent ethics review, transparency, and (where feasible) well-informed consent and debriefing [108]. Treating these interventions as ethically governed experiments, rather than as frictionless “innovation,” helps prevent manipulative or coercive uses of attention engineering.

Several governments are exploring collaborative regulatory models that combine industry self-governance with state supervision. Under such arrangements, companies participate in setting transparency and safety benchmarks while remaining subject to independent audits. This cooperative approach allows regulators to access necessary information about proprietary systems without compelling full disclosure of trade secrets. It also recognizes that no single entity—public or private—can fully monitor the evolving AI landscape. Shared governance disperses responsibility, aligning the interests of developers, policymakers, and civil society around a common goal: sustaining trustworthy attention infrastructures [104, 109, 110].

At the same time, regulators must guard against unintended concentration of power. When compliance and auditing costs are high, smaller firms, nonprofits, and academic labs may be priced out of developing or deploying advanced systems, shifting innovation and agenda-setting toward a small set of incumbents. Research on the “compute divide” in modern AI documents a growing advantage for large technology firms and elite institutions in producing state-of-the-art work as training becomes increasingly compute-intensive [111]. Competition authorities likewise warn that control over critical AI inputs (notably compute, cloud distribution, and data), combined with strategic partnerships between dominant cloud providers and frontier-model developers, can raise barriers to entry and increase lock-in [112, 113]. A concrete example is the use of equity/revenue-sharing investments paired with large, sometimes restrictive cloud-spend

commitments and information-sharing arrangements—terms that can increase switching costs and limit rivals’ access to key inputs [113]. Consequently, attention governance should encourage a pluralistic ecosystem of AI providers, academic researchers, and public-interest organizations capable of auditing algorithms and informing citizens. Transparent standards for data provenance, labeling of synthetic content, and auditing of recommendation systems will help ensure that AI enhances rather than exploits collective attention.

3.5 Toward human–AI collaboration in attention management

Ultimately, engineering collective attention is not about centralized control but about fostering collaboration between humans and intelligent systems [66]. AI has the capacity to assist humans in identifying credible information, filtering noise, and highlighting emerging issues that merit societal focus. However, to achieve these outcomes, design priorities must shift from maximizing engagement to supporting comprehension, reflection, and deliberation. Concrete examples include accuracy prompts that redirect users’ attention toward truthfulness and reduce misinformation sharing, as well as community-sourced context mechanisms (e.g., Twitter “Community Notes”), which empirical studies find can reduce diffusion and engagement once notes are attached [89, 90, 92].

Achieving collective intelligence also depends on getting incentives right. In information-rich environments, the socially optimal objective is not to maximize individual participation or raw activity, but to reward contributions that are informative and reduce correlated error (e.g., discouraging herding and encouraging diverse, independent signals). Formal work shows that appropriately designed incentives can increase collective accuracy by shifting effort toward high-impact information and away from imitative behavior [114]. This complements empirical findings that collective performance depends on group processes that broaden information use (e.g., social sensitivity and more equal participation) rather than on individual IQ alone [115]. Conversely, when social feedback and reputational pressures reward conformity, opinion diversity shrinks without improving accuracy, undermining the “wisdom of crowds” [116, 117].

The future of attention management will depend on cultivating synergy between human judgment and machine assistance. Humans contribute ethical reasoning, contextual understanding, and normative evaluation, while AI contributes scale, speed, and pattern recognition [118, 119]. When integrated responsibly, these strengths can create an information environment that amplifies knowledge rather than confusion. Examples include co-monitoring systems where AI continuously identifies potential misinformation clusters and human experts validate responses, or participatory dashboards that visualize which topics dominate public discourse, allowing citizens to consciously redirect their attention [120].

Educational institutions, news media, and civic organizations have pivotal roles in institutionalizing these collaborations. Schools can incorporate AI literacy and critical thinking into curricula, ensuring that future generations understand how automated systems influence what they see [121, 122]. Journalists can employ AI tools to track manipulation campaigns or detect coordinated inauthentic behavior, strengthening the integrity of information ecosystems [123, 124]. Governments can sponsor public-interest research and

ensure equitable access to trustworthy digital infrastructures [69, 104].

The broader vision is a society in which technology and human agency coexist in mutually reinforcing balance. Instead of being passive consumers of algorithmically curated attention, individuals become active participants in steering collective focus. Transparent systems, ethical regulation, and widespread education create conditions under which AI contributes to civic reasoning rather than distorting it.

4 Conclusion

Collective attention in the age of AI represents both an opportunity and a risk. The same technologies capable of spreading falsehoods instantaneously can also support the organization of knowledge and the coordination of social problem-solving. Whether these capacities strengthen or weaken societies depends on governance, education, and design choices.

At the individual level, promoting digital literacy and critical reasoning fortifies the human component of attention systems. At the societal level, deploying AI for verification, diversity promotion, and content labeling can redirect public focus toward verifiable and constructive topics. At the policy level, transparent standards and adaptable regulation anchor these efforts in accountability.

Improving collective attention therefore requires a holistic approach: empowering citizens, guiding institutions, and aligning AI development with ethical and epistemic goals. When combined, these strategies can transform AI from a manipulative force into a collaborative partner—one that enhances humanity's capacity to concentrate on truth, empathy, and shared understanding rather than distraction and division. Through conscious design and governance, collective attention can indeed be engineered not to control minds, but to sustain an informed and resilient global society.

Declarations

Ethical approval

Not applicable.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Data availability

Not applicable.

Code availability

Not applicable.

Ethics declaration

Not applicable.

Competing interests

The author declares no competing interests.

Acknowledgements

R.Z. gratefully acknowledges the financial support from the Australian Government Research Training Program (RTP) scholarship.

Additional information

Correspondence and requests for materials should be addressed to R.Z.

References

- [1] Garriy Shteynberg et al. “Shared worlds and shared minds: A theory of collective learning and a psychology of common knowledge.” In: *Psychological review* 127.5 (2020), p. 918.
- [2] Peter Mundy and Lisa Newell. “Attention, joint attention, and social cognition”. In: *Current directions in psychological science* 16.5 (2007), pp. 269–274.
- [3] Sophie Wohltjen and Thalia Wheatley. “Eye contact marks the rise and fall of shared attention in conversation”. In: *Proceedings of the National Academy of Sciences* 118.37 (2021), e2106645118.
- [4] Garriy Shteynberg. “Shared attention”. In: *Perspectives on psychological science* 10.5 (2015), pp. 579–590.
- [5] Erica J Boothby, Margaret S Clark, and John A Bargh. “Shared experiences are amplified”. In: *Psychological science* 25.12 (2014), pp. 2209–2216.
- [6] Garriy Shteynberg and Evan P Apfelbaum. “The power of shared experience: Simultaneous observation with similar others facilitates social learning”. In: *Social Psychological and Personality Science* 4.6 (2013), pp. 738–744.
- [7] Arianna Schiano Lomoriello et al. “Shared attention amplifies the neural processing of emotional faces”. In: *Journal of cognitive neuroscience* 34.6 (2022), pp. 917–932.
- [8] Soroush Vosoughi, Deb Roy, and Sinan Aral. “The spread of true and false news online”. In: *science* 359.6380 (2018), pp. 1146–1151.
- [9] Eytan Bakshy, Solomon Messing, and Lada A Adamic. “Exposure to ideologically diverse news and opinion on Facebook”. In: *Science* 348.6239 (2015), pp. 1130–1132.

- [10] Michela Del Vicario et al. “The spreading of misinformation online”. In: *Proceedings of the National Academy of Sciences* 113.3 (2016), pp. 554–559. DOI: <https://doi.org/10.1073/pnas.1517441113>.
- [11] David M. J. Lazer et al. “The science of fake news”. In: *Science* 359.6380 (2018), pp. 1094–1096. DOI: [10.1126/science.aao2998](https://doi.org/10.1126/science.aao2998).
- [12] Adrien Friggeri et al. “Rumor Cascades”. In: *Proceedings of the International AAAI Conference on Web and Social Media*. Vol. 8. 1. AAAI, 2014, pp. 101–110. DOI: <https://doi.org/10.1609/icwsm.v8i1.14559>.
- [13] Matteo Cinelli et al. “The echo chamber effect on social media”. In: *Proceedings of the national academy of sciences* 118.9 (2021), e2023301118.
- [14] William J Brady et al. “Emotion shapes the diffusion of moralized content in social networks”. In: *Proceedings of the National Academy of Sciences* 114.28 (2017), pp. 7313–7318.
- [15] Rui Fan et al. “Anger is more influential than joy: Sentiment correlation in Weibo”. In: *PloS one* 9.10 (2014), e110184.
- [16] William J Brady et al. “How social learning amplifies moral outrage expression in online social networks”. In: *Science Advances* 7.33 (2021), eabe5641.
- [17] Herbert A. Simon. “Designing Organizations for an Information-Rich World”. In: *Computers, Communications, and the Public Interest*. Ed. by Martin Greenberger. Baltimore: The Johns Hopkins Press, 1971, pp. 37–72.
- [18] Philipp Lorenz-Spreen et al. “Accelerating dynamics of collective attention”. In: *Nature communications* 10.1 (2019), p. 1759.
- [19] Roger E Kasperson et al. “The social amplification of risk: A conceptual framework”. In: *Risk analysis* 8.2 (1988), pp. 177–187.
- [20] Anthony Downs. “Up and Down with Ecology-the Issue-Attention Cycle”. In: *The Public Interest* 28 (1972), p. 38.
- [21] Frank R Baumgartner and Bryan D Jones. “Agenda dynamics and policy subsystems”. In: *The journal of Politics* 53.4 (1991), pp. 1044–1074.
- [22] David Lazer et al. “The parable of Google Flu: traps in big data analysis”. In: *science* 343.6176 (2014), pp. 1203–1205.
- [23] Fang Wu and Bernardo A Huberman. “Novelty and collective attention”. In: *Proceedings of the National Academy of Sciences* 104.45 (2007), pp. 17599–17601.
- [24] Allison JB Chaney, Brandon M Stewart, and Barbara E Engelhardt. “How algorithmic confounding in recommendation systems increases homogeneity and decreases utility”. In: *Proceedings of the 12th ACM conference on recommender systems*. 2018, pp. 224–232.

- [25] Ferenc Huszár et al. “Algorithmic amplification of politics on Twitter”. In: *Proceedings of the national academy of sciences* 119.1 (2022), e2025334119.
- [26] Paul Covington, Jay Adams, and Emre Sargin. “Deep Neural Networks for YouTube Recommendations”. In: *Proceedings of the 10th ACM Conference on Recommender Systems (RecSys '16)*. Association for Computing Machinery, 2016, pp. 191–198. DOI: [10.1145/2959100.2959190](https://doi.org/10.1145/2959100.2959190).
- [27] Abhijit V Banerjee. “A simple model of herd behavior”. In: *The quarterly journal of economics* 107.3 (1992), pp. 797–817.
- [28] Sushil Bikhchandani, David Hirshleifer, and Ivo Welch. “A theory of fads, fashion, custom, and cultural change as informational cascades”. In: *Journal of political Economy* 100.5 (1992), pp. 992–1026.
- [29] David S Scharfstein and Jeremy C Stein. “Herd behavior and investment”. In: *The American economic review* (1990), pp. 465–479.
- [30] Zhi Da, Joseph Engelberg, and Pengjie Gao. “In search of attention”. In: *The journal of finance* 66.5 (2011), pp. 1461–1499.
- [31] Brad M Barber and Terrance Odean. “All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors”. In: *The review of financial studies* 21.2 (2008), pp. 785–818.
- [32] Walter Lippmann. *Public opinion*. Routledge, 2017.
- [33] Harold D Lasswell. *Propaganda technique in world war I*. MIT press, 1971.
- [34] Edward L Bernays. “The engineering of consent”. In: *The Annals of the American Academy of Political and Social Science* 250.1 (1947), pp. 113–120.
- [35] Jacques Ellul. *Propaganda: The formation of men’s attitudes*. Vintage, 2021.
- [36] Charles T Salmon. “Manufacturing consent: The political economy of the mass media”. In: *Journalism and Mass Communication Quarterly* 66.2 (1989), p. 494.
- [37] Karen Yeung. ““Hypernudge”: Big Data as a Mode of Regulation by Design”. In: *Information, Communication & Society* 20.1 (2017), pp. 118–136. DOI: <https://doi.org/10.1080/1369118X.2016.1186713>.
- [38] Frederik J. Zuiderveen Borgesius et al. “Online Political Microtargeting: Promises and Threats for Democracy”. In: *Utrecht Law Review* 14.1 (2018), pp. 82–96. DOI: [10.18352/ulr.420](https://doi.org/10.18352/ulr.420).
- [39] Dirk Helbing, ed. *Towards Digital Enlightenment: Essays on the Dark and Light Sides of the Digital Revolution*. Springer, 2019. DOI: <https://doi.org/10.1007/978-3-319-90869-4>.

- [40] Shoshana Zuboff. *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*. New York: PublicAffairs, 2019.
- [41] Y. Ma, X. Zhang, J. Ren, et al. “Linguistic features of AI mis/disinformation and the detection limits of LLMs”. In: *Nature Communications* 17 (2026), p. 456. DOI: <https://doi.org/10.1038/s41467-025-67145-1>.
- [42] Angus R. Williams et al. *Large language models can consistently generate high-quality content for election disinformation operations*. Aug. 13, 2024. arXiv: 2408.06731. URL: <https://arxiv.org/abs/2408.06731> (visited on 02/07/2026).
- [43] Donghee Shin. *Artificial Misinformation: Exploring Human-Algorithm Interaction Online*. Palgrave Macmillan Cham, 2024. DOI: <https://doi.org/10.1007/978-3-031-52569-8>.
- [44] Lukasz Olejnik, Minh-Dung Tran, and Claude Castelluccia. “Selling Off Privacy at Auction”. In: *Proceedings of the Network and Distributed System Security Symposium (NDSS)*. 2014.
- [45] Michael Veale and Frederik Zuiderveen Borgesius. “Adtech and real-time bidding under European data protection law”. In: *German Law Journal* 23.2 (2022), pp. 226–256.
- [46] US Senate. “Report on Russian Active Measures Campaigns and Interference in the 2016 US Election, Volume 2: Russia’s Use of Social Media with Additional Views”. In: *Washington, DC: United States Senate Select Committee on Intelligence*. https://www.intelligence.senate.gov/sites/default/files/documents/Report_Volume2.pdf (2019).
- [47] Karthika Subramani et al. “When push comes to ads: Measuring the rise of (malicious) push advertising”. In: *Proceedings of the ACM Internet Measurement Conference*. 2020, pp. 724–737.
- [48] Brett Stone-Gross et al. “Understanding fraudulent activities in online ad exchanges”. In: *Proceedings of the 2011 ACM SIGCOMM conference on Internet measurement conference*. 2011, pp. 279–294.
- [49] Smitha Milli et al. “Engagement, user satisfaction, and the amplification of divisive content on social media”. In: *PNAS nexus* 4.3 (2025), pgaf062.
- [50] William J Brady et al. “Algorithm-mediated social learning in online social networks”. In: *Trends in Cognitive Sciences* 27.10 (2023), pp. 947–960.
- [51] Lilian Weng et al. “Competition among memes in a world with limited attention”. In: *Scientific reports* 2.1 (2012), p. 335.
- [52] Xiaoyan Qiu et al. “Limited individual attention and online virality of low-quality information”. In: *Nature Human Behaviour* 1.7 (2017), p. 0132.
- [53] Tim Wu. *The attention merchants: The epic scramble to get inside our heads*. Vintage, 2017.

- [54] David Lazer and Allan Friedman. “The network structure of exploration and exploitation”. In: *Administrative science quarterly* 52.4 (2007), pp. 667–694.
- [55] Ray Jiang et al. “Degenerate feedback loops in recommender systems”. In: *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society*. 2019, pp. 383–390.
- [56] Daniel Fleder and Kartik Hosanagar. “Blockbuster Culture’s Next Rise or Fall: The Impact of Recommender Systems on Sales Diversity”. In: *Management Science* 55.5 (2009), pp. 697–712. DOI: [10.1287/mnsc.1080.0974](https://doi.org/10.1287/mnsc.1080.0974). eprint: <https://doi.org/10.1287/mnsc.1080.0974>. URL: <https://doi.org/10.1287/mnsc.1080.0974>.
- [57] Anastasiia Klimashevskaya et al. “A survey on popularity bias in recommender systems”. In: *User Modeling and User-Adapted Interaction* 34.5 (2024), pp. 1777–1834.
- [58] Juan Perdomo et al. “Performative Prediction”. In: *Proceedings of the 37th International Conference on Machine Learning*. Ed. by Hal Daumé III and Aarti Singh. Vol. 119. Proceedings of Machine Learning Research. PMLR, July 2020, pp. 7599–7609.
- [59] Ilia Shumailov et al. “AI models collapse when trained on recursively generated data”. In: *Nature* 631.8022 (2024), pp. 755–759.
- [60] Ilia Shumailov et al. “The Curse of Recursion: Training on Generated Data Makes Models Forget”. In: *arXiv* (2023). arXiv:2305.17493. DOI: [10.48550/arXiv.2305.17493](https://doi.org/10.48550/arXiv.2305.17493).
- [61] Sina Alemohammad et al. “Self-consuming generative models go mad”. In: *The Twelfth International Conference on Learning Representations*. 2023.
- [62] Philip M Napoli. *Social media and the public interest: Media regulation in the disinformation age*. Columbia university press, 2019.
- [63] Engin Bozdog. “Bias in algorithmic filtering and personalization”. In: *Ethics and information technology* 15.3 (2013), pp. 209–227.
- [64] Rishi Bommasani. “On the opportunities and risks of foundation models”. In: *arXiv preprint arXiv:2108.07258* (2021).
- [65] Emily M. Bender et al. “On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?” In: *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*. Association for Computing Machinery, 2021, pp. 610–623. DOI: <https://doi.org/10.1145/3442188.3445922>.
- [66] Donghee Shin. *Debiasing AI: Rethinking the intersection of innovation and sustainability*. Routledge, 2025.
- [67] OECD. “Scoping the OECD AI principles: Deliberations of the Expert Group on Artificial Intelligence at the OECD (AIGO)”. In: *OECD Digital Economy Papers* 291 (2019).

- [68] UNESCO. *Recommendation on the Ethics of Artificial Intelligence*. Nov. 23, 2021. URL: <https://www.unesco.org/en/legal-affairs/recommendation-ethics-artificial-intelligence> (visited on 02/06/2026).
- [69] National Institute of Standards and Technology. *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. Tech. rep. NIST AI 100-1. National Institute of Standards and Technology, 2023. DOI: [10.6028/NIST.AI.100-1](https://doi.org/10.6028/NIST.AI.100-1). URL: <https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf> (visited on 02/06/2026).
- [70] White House Office of Science and Technology Policy. *What is the Blueprint for an AI Bill of Rights?* Oct. 4, 2022. URL: <https://bidenwhitehouse.archives.gov/ostp/ai-bill-of-rights/what-is-the-blueprint-for-an-ai-bill-of-rights/> (visited on 02/06/2026).
- [71] Laura Weidinger et al. “Taxonomy of Risks Posed by Language Models”. In: *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. 2022. DOI: <https://doi.org/10.1145/3531146.3533088>.
- [72] Donghee Shin. “How do people judge the credibility of algorithmic sources?” In: *Ai & Society* 37.1 (2022), pp. 81–96.
- [73] Philipp Lorenz-Spreen et al. “A systematic review of worldwide causal and correlational evidence on digital media and democracy”. In: *Nature human behaviour* 7.1 (2023), pp. 74–101.
- [74] Yisroel Mirsky and Wenke Lee. “The Creation and Detection of Deepfakes: A Survey”. In: *ACM Computing Surveys* 54.1 (2021), pp. 1–41. DOI: <https://doi.org/10.1145/3425780>.
- [75] Mateusz Łabuz and Christopher Nehring. “On the way to deep fake democracy? Deep fakes in election campaigns in 2023”. In: *European Political Science* 23 (2024), pp. 454–473. DOI: <https://doi.org/10.1057/s41304-024-00482-9>.
- [76] Robert Chesney and Danielle Keats Citron. “Deep Fakes: A Looming Challenge for Privacy, Democracy, and National Security”. In: *California Law Review* 107.6 (2019), pp. 1753–1819.
- [77] Emilio Ferrara et al. “The Rise of Social Bots”. In: *Communications of the ACM* 59.7 (2016), pp. 96–104. DOI: <https://doi.org/10.1145/2818717>.
- [78] Chengcheng Shao et al. “The spread of low-credibility content by social bots”. In: *Nature Communications* 9 (2018), p. 4787. DOI: <https://doi.org/10.1038/s41467-018-06930-7>.
- [79] Onur Varol et al. “Online Human-Bot Interactions: Detection, Estimation, and Characterization”. In: *Proceedings of the Eleventh International AAAI Conference on Web and Social Media (ICWSM 2017)*. 2017, pp. 280–289.

- [80] Stefano Cresci et al. “The Paradigm-Shift of Social Spambots: Evidence, Theories, and Tools for the Arms Race”. In: *Proceedings of the 26th International Conference on World Wide Web Companion (WWW '17 Companion)*. International World Wide Web Conferences Steering Committee, 2017, pp. 963–972. DOI: <https://doi.org/10.1145/3041021.3055135>.
- [81] Melisa Basol, Jon Roozenbeek, and Sander van der Linden. “Good News about Bad News: Gamified Inoculation Boosts Confidence and Cognitive Immunity Against Fake News”. In: *Journal of Cognition* 3.1 (2020), p. 2. DOI: [10.5334/joc.91](https://doi.org/10.5334/joc.91). URL: <https://journalofcognition.org/en/articles/10.5334/joc.91>.
- [82] Jon Roozenbeek and Sander van der Linden. “Fake news game confers psychological resistance against online misinformation”. In: *Humanities and Social Sciences Communications* 5 (2019), p. 65. DOI: <https://doi.org/10.1057/s41599-019-0279-9>. URL: <https://www.nature.com/articles/s41599-019-0279-9>.
- [83] Melisa Basol et al. “Towards psychological herd immunity: Cross-cultural evidence for two pre-bunking interventions against COVID-19 misinformation”. In: *Big Data & Society* 8.1 (2021). DOI: <https://doi.org/10.1177/20539517211013868>. URL: <https://journals.sagepub.com/doi/10.1177/20539517211013868> (visited on 02/06/2026).
- [84] J. E. Brodsky et al. “Improving college students’ fact-checking strategies through lateral reading instruction in a general education civics course”. In: *Cognitive Research: Principles and Implications* 6 (2021), p. 23. DOI: <https://doi.org/10.1186/s41235-021-00291-4>. URL: <https://cognitiveresearchjournal.springeropen.com/articles/10.1186/s41235-021-00291-4>.
- [85] Russell G Smith and Tabor Akman. “Raising public awareness of consumer fraud in Australia.” In: *Trends & Issues in Crime & Criminal Justice* 349 (2008).
- [86] Rasmus Ingemann Tuffveson Jensen, Julie Gerlings, and Joras Ferwerda. “Do Awareness Campaigns Reduce Financial Fraud? RIT Jensen et al.” In: *European Journal on Criminal Policy and Research* 31.4 (2025), pp. 717–752.
- [87] Justin Sherman, D. A. Stokes, Elissa M. Redmiles, et al. “Designing Media Provenance Indicators to Combat Fake Media”. In: *Proceedings of the 24th International Symposium on Research in Attacks, Intrusions and Defenses (RAID 2021)*. 2021. DOI: <https://doi.org/10.1145/3471621.3471860>.
- [88] Jonathan Stray et al. “What are you optimizing for? aligning recommender systems with human values”. In: *arXiv preprint arXiv:2107.10939* (2021).
- [89] Isaac Slaughter et al. “Community notes moderate engagement with and diffusion of false information online”. In: *arXiv preprint arXiv:2502.13322* (2025). URL: <https://arxiv.org/abs/2502.13322>.

- [90] Thomas Renault, David Restrepo Amariles, and Aurore Troussel. “Collaboratively adding context to social media posts reduces the sharing of false news”. In: *arXiv preprint arXiv:2404.02803* (2024). URL: <https://arxiv.org/abs/2404.02803>.
- [91] Cameron Martel and David G Rand. “Misinformation warning labels are widely effective: A review of warning effects and their moderating features”. In: *Current Opinion in Psychology* 54 (2023), p. 101710. DOI: <https://doi.org/10.1016/j.copsy.2023.101710>.
- [92] Gordon Pennycook et al. “Shifting attention to accuracy can reduce misinformation online”. In: *Nature* 592.7855 (2021), pp. 590–595. DOI: <https://doi.org/10.1038/s41586-021-03344-2>.
- [93] Cameron Martel and David G Rand. “Fact-checker warning labels are effective even for those who distrust fact-checkers”. In: *Nature Human Behaviour* 8.10 (2024), pp. 1957–1967.
- [94] Natali Helberger, Kari Karppinen, and Lucia D’acunto. “Exposure diversity as a design principle for recommender systems”. In: *Information, communication & society* 21.2 (2018), pp. 191–207.
- [95] Przemysław Kazienko, Erik Cambria, et al. “Toward Responsible Recommender Systems”. In: *IEEE Intelligent Systems* (2024). DOI: <https://doi.org/10.1109/MIS.2024.3398190>.
- [96] A. Kozyreva, S. Lewandowsky, and R. Hertwig. “Citizens Versus the Internet: Confronting Digital Challenges With Cognitive Tools”. In: *Psychological Science in the Public Interest* 21 (2019), pp. 103–156. DOI: <https://doi.org/10.1177/1529100620946707>.
- [97] Joseph B. Bak-Coleman et al. “Combining interventions to reduce the spread of viral misinformation”. In: *Nature Human Behaviour* 6.9 (2022), pp. 1236–1245. DOI: <https://doi.org/10.1038/s41562-022-01388-6>.
- [98] Cyberspace Administration of China. *Measures for the Identification of Artificial Intelligence-Generated and Synthetic Content*. In force from 2025-09-01 (China). Mar. 14, 2025. URL: https://www.cac.gov.cn/2025-03/14/c_1743654684782215.htm (visited on 02/06/2026).
- [99] European Commission. *Artificial Intelligence Act (Regulation (EU) 2024/1689): Article 50 (Transparency obligations)*. June 13, 2024. URL: <https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-50> (visited on 02/06/2026).
- [100] California State Legislature. *California Assembly Bill No. 2355 (2023–2024): Political Reform Act of 1974: political advertisements: artificial intelligence (Chapter 260, Statutes of 2024)*. Sept. 17, 2024. URL: <https://legiscan.com/CA/bill/AB2355/2023> (visited on 02/06/2026).
- [101] Nicholas Diakopoulos. “Accountability in Algorithmic Decision Making”. In: *Communications of the ACM* 59.2 (2016), pp. 56–62. DOI: <https://doi.org/10.1145/2844110>.

- [102] Tarleton Gillespie. *Custodians of the Internet: Platforms, content moderation, and the hidden decisions that shape social media*. Yale University Press, 2018.
- [103] Jenna Burrell. “How the machine ‘thinks’: Understanding opacity in machine learning algorithms”. In: *Big Data & Society* 3.1 (2016). DOI: [10.1177/2053951715622512](https://doi.org/10.1177/2053951715622512).
- [104] Gregory Falco et al. “Governing AI safety through independent audits”. In: *Nature Machine Intelligence* 3.7 (2021), pp. 566–571. DOI: <https://doi.org/10.1038/s42256-021-00370-7>.
- [105] Council of Europe. *Council of Europe Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law (CETS No. 225)*. Sept. 5, 2024. URL: <https://book.coe.int/en/texts-of-council-of-europe-treaties/12225-council-of-europe-framework-convention-on-artificial-intelligence-and-human-rights-democracy-and-the-rule-of-law-cets-no-225.html> (visited on 02/06/2026).
- [106] European Commission. *Hiroshima Process International Guiding Principles for Advanced AI System*. Oct. 30, 2023. URL: <https://digital-strategy.ec.europa.eu/en/library/hiroshima-process-international-guiding-principles-advanced-ai-system> (visited on 02/06/2026).
- [107] Executive Office of the President. *Executive Order 14110: Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence (Federal Register publication)*. Federal Register, Vol. 88, No. 210. Nov. 1, 2023. URL: <https://www.govinfo.gov/content/pkg/FR-2023-11-01/html/2023-24283.htm> (visited on 02/06/2026).
- [108] Verena Zimmermann. “Smart cities as a testbed for experimenting with humans?-Applying psychological ethical guidelines to smart city interventions”. In: *Ethics and Information Technology* 25.4 (2023), p. 54.
- [109] Neil Gunningham and Joseph Rees. “Industry Self-Regulation: An Institutional Perspective”. In: *Law & Policy* 19.4 (1997), pp. 363–414. DOI: <https://doi.org/10.1111/1467-9930.t01-1-00033>.
- [110] Peter M. DeMarzo, Michael J. Fishman, and Kathleen M. Hagerty. “Self-Regulation and Government Oversight”. In: *The Review of Economic Studies* 72.3 (2005), pp. 687–706. DOI: <https://doi.org/10.1111/j.1467-937X.2005.00348.x>.
- [111] Nur Ahmed and Muntasir Wahed. “The De-democratization of AI: Deep learning and the compute divide in artificial intelligence research”. In: *arXiv preprint arXiv:2010.15581* (2020).
- [112] Competition and Markets Authority. *AI Foundation Models: Update paper*. Apr. 11, 2024. URL: <https://www.gov.uk/government/publications/ai-foundation-models-update-paper> (visited on 02/07/2026).

- [113] Federal Trade Commission. *FTC issues staff report on AI partnerships & investments study*. Jan. 17, 2025. URL: <https://www.ftc.gov/news-events/news/press-releases/2025/01/ftc-issues-staff-report-ai-partnerships-investments-study> (visited on 02/07/2026).
- [114] Rory P. Mann and Dirk Helbing. “Optimal incentives for collective intelligence”. In: *Proceedings of the National Academy of Sciences* 114.20 (2017), pp. 5077–5082. DOI: <https://doi.org/10.1073/pnas.1618722114>.
- [115] Anita Williams Woolley et al. “Evidence for a collective intelligence factor in the performance of human groups”. In: *Science* 330.6004 (2010), pp. 686–688. DOI: [10.1126/science.1193147](https://doi.org/10.1126/science.1193147).
- [116] Jan Lorenz et al. “How social influence can undermine the wisdom of crowd effect”. In: *Proceedings of the national academy of sciences* 108.22 (2011), pp. 9020–9025.
- [117] Lev Muchnik, Sinan Aral, and Sean J Taylor. “Social influence bias: A randomized experiment”. In: *Science* 341.6146 (2013), pp. 647–651.
- [118] Ece Kamar. “Directions in Hybrid Intelligence: Complementing AI Systems with Human Intelligence”. In: *Proceedings of the Twenty-Fifth International Joint Conference on Artificial Intelligence (IJCAI-16)*. IJCAI, 2016, pp. 4070–4073.
- [119] Saleema Amershi et al. “Guidelines for Human–AI Interaction”. In: *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, 2019, pp. 1–13. DOI: <https://doi.org/10.1145/3290605.3300233>.
- [120] Dominik Sacha et al. “What You See Is What You Can Change: Human-Centred Machine Learning by Interactive Visualization”. In: *Neurocomputing* 268 (2017), pp. 164–175. DOI: <https://doi.org/10.1016/j.neucom.2017.01.105>.
- [121] Duri Long and Brian Magerko. “What is AI Literacy? Competencies and Design Considerations”. In: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, 2020, pp. 1–16. DOI: <https://doi.org/10.1145/3313831.3376727>.
- [122] Sarah McGrew. “Learning to evaluate: An intervention in civic online reasoning”. In: *Computers & Education* 145 (2020), p. 103711. DOI: <https://doi.org/10.1016/j.compedu.2019.103711>.
- [123] Zhijiang Guo, Michael Schlichtkrull, and Andreas Vlachos. “A Survey on Automated Fact-Checking”. In: *Transactions of the Association for Computational Linguistics* 10 (2022), pp. 178–206. DOI: https://doi.org/10.1162/tacl_a_00454.

- [124] Serena Tardelli et al. “Multifaceted online coordinated behavior in the 2020 US presidential election”. In: *EPJ Data Science* 13.33 (2024). DOI: <https://doi.org/10.1140/epjds/s13688-024-00467-0>.

ARTICLE IN PRESS